GROUP MEMBERS:

1. SCM223-0222/2024 RAEL JUMA-GROUP LEADER

2. SCM223-1457/2024 FAITH ZUMA

3. SCM223-0205/2024 JACINTA MUTINDA

4. SCM223-1580/2024 IAN KIBERU

5. SCM223-1431/2024 BENTA YVONNE

6. SCM223-1454/2024 FESTUS MULEI

7. SCM223-0238/2024 VANESSA KERUBO

8. SCM223-1358/2024 IAN GITAU

STA 2240:FUNDAMENTALS OF OOP

STATISTICS 2.2 B

ASSIGNMENT (MILESTONE 5 & 6) SUMMARY

Milestone 5 Summary

It focuses on transforming the Steam Games Data Cleaning & Transformation Pipeline into a concurrent and high-performance system.

The system implemented multithreading, multiprocessing, asynchronous programming, task queues, and parallel processing pipelines to improve efficiency and scalability.

Thread safety mechanisms were also introduced to prevent race conditions during concurrent execution. Performance benchmarking was used to compare execution speed before and after optimization.

Implementing **Multiprocessing** to clean the dataset 4x faster by using all CPU cores. Using **AsyncIO** to fetch missing metadata from external web APIs.

It leads to a high-speed system capable of handling "Big Data" volumes.

Milestone 6 Summary

It focuses on integrating all system components into a research-level intelligent data processing system.

The system introduced automated data quality scoring, intelligent cleaning strategy selection, automated validation, and integrated processing pipelines.

The final architecture combined object-oriented programming, concurrency, functional programming, exception handling, and intelligent automation into a scalable and reliable Steam dataset processing platform.

It results to a sophisticated, intelligent research tool that provides deep market insights.